

Simultaneous determination of aloin A and aloe emodin in products containing *Aloe vera* by ultra-performance liquid chromatography with tandem mass spectrometry

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A novel method for the simultaneous determination of aloin A and aloe emodin in products labeled to contain *Aloe vera* has been developed and validated. This analytical procedure involved extracting a test portion of the product in acetonitrile : water (40 : 60, v/v), followed by centrifugation and filtration through a 0.2 μm PTFE filter. Quantitation was achieved using ultra-performance liquid chromatography-tandem mass spectrometry (UPLC-MS/MS) with electrospray ionization (ESI) in the negative mode. The separation was carried out using a Waters Acquity UPLC BEH C_{18} column; 2.1 mm $\times$ 50 mm; 1.7 μm particle size. A five-minute gradient program was optimized at a flow rate of 0.5 mL min⁻¹. Both water with 0.5% acetic acid and methanol with 0.5% acetic acid were used as mobile phases. The linear range was from 5.0 to 1000 ng mL⁻¹ for aloin A and from 1.0 to 500 ng mL⁻¹ for aloe emodin, with $r \geq 0.995$ for both analytes. Since serious matrix effects (both ion suppression and ion enhancement) were eliminated by dilution for all samples, the recoveries for aloin A and aloe emodin ranged from 89–118%, and 74–108%, respectively, in the matrices tested. The method has been successfully applied to various products labeled to contain *Aloe vera* including liquids, capsules, soft gels and tabs. The amount of aloin A and aloe emodin in the samples ranged from none detected to 160.0 mg g⁻¹ and from none detected to 3.8 mg g⁻¹, respectively.

Introduction

Aloe vera, also known as *Aloe barbadensis* Miller, is one of over 400 species of flowering, succulent plants in the genus *Aloe*. *Aloe vera* is a stemless or short-stemmed plant which grows in regions having hot and dry climates including Africa, South Asia and the southern United States.¹ *Aloe vera* has long been used in traditional medicines in countries such as India, China and Japan.² Currently, *Aloe vera* is widely used as an ingredient in food, beverages, dietary supplements, cosmetics and pharmaceuticals due to its aromatic properties, bitter taste, and pharmacological activities. Pharmacological properties attributed to *Aloe vera* include antiproliferative and antitumor activities, wound healing, anti-inflammatory effects, and purgative action.^{1–3}

The leaf of *Aloe vera* comprises an outer rind or cuticle and an inner pulp, which differ significantly in chemical composition. Aloe latex, derived from the rind of *Aloe vera* leaves, is a bitter tasting liquid containing a variety of phenolic compounds.⁴ These include aloin A (Fig. 1A), also known as barbaloin. Aloin A, an anthraquinone-C glycoside, has been shown to be responsible for the purgative activity of *Aloe vera*.¹ Both aloin A (configuration at

C10, C1: S,S) and its diastereoisomer, aloin B (C10, C1: R,S), are found in aloe latex⁴ and in the commercial product aloin, derived from *Aloe vera* latex. Owing to its pharmacological activity, the amount of aloin used as a flavoring agent in food products is restricted by the European Economic Community (EEC) Council Directive 88/388.⁵ Other phenolic compounds found in aloe latex include aloesin, aloenin and aloe emodin.⁴ The structure of aloe emodin is given in Fig. 1B. Aloe emodin, present in aloe latex in small amounts relative to aloin A and aloin B, can be formed by oxidative cleavage of the glycosidic linkage of aloin A or aloin B. Aloe emodin, a metabolite produced from aloin A or aloin B by intestinal bacteria, exerts synergistic purgative effects with its

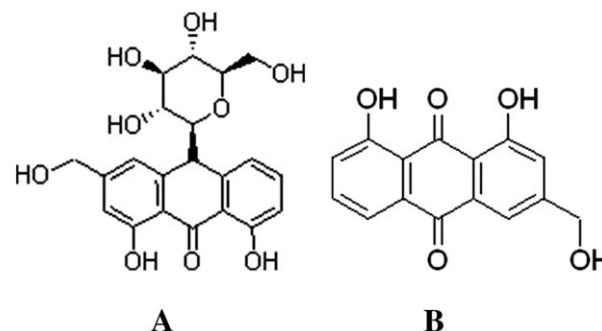


Fig. 1 Chemical structures of (A) aloin A and (B) aloe emodin.

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parent compounds.¹ The inner pulp of the aloe leaf contains aloe gel. Aloe gel contains more than 98% water. The carbohydrate acemannan accounts for greater than 60% of the solid matter found in aloe gel.^{3,6,7} The immunostimulating activity reported for aloe gel has been attributed to acemannan. Free monosaccharides represent another major constituent in aloe gel.³

Several general methods are used to produce the *Aloe vera* found in consumer products. The whole leaf extract or juice is prepared by grinding whole leaves, frequently following treatment with cellulase to digest the rind of the leaf. To obtain a product devoid of phenolic compounds (including anthraquinones and their glycosides), the resulting slurry may be filtered through charcoal.⁸ *Aloe vera* inner leaf juice or gel is prepared by removing the rind, either mechanically or by hand, prior to processing. Any remaining latex is then rinsed away. The remaining gelatinous inner-leaf material is then ground or crushed.⁸ It is important that the ingredient used in a consumer product is appropriate for the product's intended use.

For products intended for oral consumption, the International Aloe Science Council (IASC) recommends the aloin in products to be less than 10 ppm.⁹ Since aloin can be oxidized into aloe emodin,¹ this study focused on both aloin A and aloe emodin.

Different analytical methods have been developed to study and determine aloin A and aloe emodin. Synchronous fluorescence spectroscopy was used to analyze aloe emodin in simulated mixtures.¹⁰ High performance thin layer chromatography (HPTLC)¹¹ and thin-layer chromatography (TLC) systems were used to analyze the root components and leaf constituents.^{12–15} The application of capillary electrophoresis to analyze anthraquinones was reported.^{16–20} Capillary gas chromatography coupled with flame ionization and mass spectrometric detection was used to determine anthraquinones in radix *Polygoni multiflori*.²¹ The most widely used techniques to analyze aloin A and aloe emodin are high performance liquid chromatography (HPLC) with UV/PDA,^{4,22–35} fluorescence^{36,37} or mass selective detectors (MSD).^{38–40} Ultra-performance liquid chromatography (UPLC)^{41–43} and nano LC methods⁴⁴ have also been used to analyze anthraquinone derivatives.

Some of the above methods exhibit drawbacks such as longer analysis times (more than 20 minutes (ref. 38, 40 and 44)), higher detection limits and lack of confirmation of the identity of analytes. The objective of this study was to develop a rapid method using ultra-performance liquid chromatography and tandem mass spectrometry (UPLC-MS/MS) detection for aloe products. This method has a short chromatographic analysis time (5.0 minutes) and low detection limits. The method has been evaluated and successfully applied to commercial products. This is the first UPLC-MS/MS method for the determination of aloin A and aloe emodin in products containing *Aloe vera* that did not require extensive sample cleanup. Since each product varied with regard to matrix, the matrix effects were evaluated by first screening every product for recovery prior to final quantitation.

Materials and methods

Chemicals and materials

LC-MS grade methanol (MeOH) and water were purchased from Fisher Scientific (Fair Lawn, NJ, USA). Acetic acid

(glacial) was purchased from Mallinckrodt Baker (Phillipsburg, NJ). Dimethyl sulfoxide (DMSO), aloin A and aloe emodin were purchased from Sigma-Aldrich (St. Louis, MO). Nitrogen gas was provided by an LC/MS Gas Generator (Source SF 5000, Parker Balson, Haverhill, MA). Fourteen commercial products containing *Aloe vera* were selected and purchased from the internet in different forms including liquids, tabs, capsules and soft gels. Products that did not contain the analytes of interest were also selected based on their label information indicating that the processing of *Aloe vera* in those products had removed aloe emodin and aloin. The ingredients/compositions of the fourteen aloe products are summarized in Table 1.

Instrumentation

Chromatographic studies were performed via a Waters ACQUITY UPLC® (Waters, Milford, MA) consisting of a binary

Table 1 Commercial aloe products^a

Product #	Form	Composition (from product label)
1	Liq.	Purified water, <i>Aloe vera</i> , 35% food grade hydrogen peroxide, colloidal silver (8 ppm), <i>Stevia</i> .
2	Liq.	<i>Aloe vera</i> , water, xanthan gum, citric acid, pure food grade sodium benzoate and potassium sorbate.
3	Liq.	<i>Aloe barbadensis</i> Miller, 100% fractionally distilled.
4	Liq.	<i>Aloe barbadensis</i> Miller whole leaf juice concentrate (92.5%), Azteca trace minerals, natural cherry and cranberry juice, citric acid, potassium sorbate, and sodium benzoate. Contains allowable amounts of the valuable anthraquinone complex: aloin and aloe emodin (4 ppm).
5	Liq.	Certified organic <i>Aloe vera</i> leaves/juice, citric acid, potassium sorbate.
6	Liq.	<i>Aloe vera</i> leaf, Cat's Claw bark extract, Chamomile flower extract, Burdock root extract, Hawthorn Berry extract, <i>Astragalus</i> root extract, Sheep Sorrel herb extract, Pau d'Arco bark extract, Slippery Elm bark extract, Rhubarb root extract, colloidal silver (water and silver, 2 ppm), sodium benzoate.
7	Liq.	<i>Aloe vera</i> concentrate (40 : 1 concentrate), potassium sorbate, citric acid, ascorbic acid, sodium benzoate.
8	Cap.	<i>Aloe barbadensis</i> (inner leaf gel, 166 mg, 200 : 1 concentrate), vegetable cellulose capsule. Each capsule is equivalent to 1 oz of fresh, mature aloe inner leaf gel when reconstituted.
9	Cap.	<i>Aloe vera</i> , whole leaf (400 mg).
10	Cap.	<i>Aloe ferox</i> (latex, 200 mg), beet (root, 150 mg), fennel (seed, 150 mg), <i>Aloe barbadensis</i> (leaf, 50 mg), plant-derived capsule, magnesium stearate.
11	Cap.	<i>Aloe ferox</i> (775 mg), plant cellulose (Vcaps).
12	S. gel	<i>Aloe vera</i> extract (200 : 1 concentrate, 75 mg), rice bran oil, soft gel capsule (gelatin, glycerin, water), beeswax, silica.
13	S. gel	<i>Aloe barbadensis</i> extract (200 : 1 concentrate, 75 mg), rice bran oil, soft gel capsule (gelatin, glycerin, water), beeswax, silica.
14	Tab	<i>Aloe vera</i> extract (6 mg), starch, cellulose, water, xylitol, aspartame, sorbitol, glycerol, malic acid, sucrose fatty acid esters.

^a Note: Liq. = liquid; Cap. = capsule; S. gel = soft gel; Tab = thin sheets approximately 2 cm × 3 cm.

solvent manager and sample manager. The mass spectrometer was an AB Sciex QTRAP 5500 (AB Sciex, Foster City, CA) equipped with a Turbo Ion Spray™ source (ESI).

Chromatographic conditions

The temperature of the sample manager was set to 5 °C during operation. The separation was carried out using a Waters ACQUITY UPLC BEH C₁₈ column; 2.1 mm × 50 mm; 1.7 μm particle size. A five-minute gradient program was optimized at a flow rate of 0.5 mL min⁻¹ with an injection volume of 5 μL. The mobile phase A was composed of 0.5% acetic acid in water (v/v) and the mobile phase B was composed of 0.5% acetic acid in methanol (v/v). The gradient parameters were: 0–0.40 min 40% B, 0.40–2.0 min from 40% B to 85% B, 2.0–3.2 min 85% B, 3.2–3.6 min from 85% to 100% B, 3.6–4.0 min 100% B, 4.0–4.4 min from 100% B to 40% B, 4.4–5.0 min 40% B. The LC eluent was introduced directly into the electrospray ionization source without splitting the flow of the eluent.

MS settings

The UPLC-ESI-MS/MS data were acquired and processed by AB Sciex Analyst software version 1.5.1. The ESI voltage was set at –4500 V in the negative ion mode. The turbo heater was maintained at 400 °C. Nitrogen was used as the curtain gas and collision gas. The curtain gas, ion source gas 1 and ion source gas 2 were set at pressures of 40, 50 and 50 PSI, respectively. The collision gas was set at “medium”. The mass spectrometer was used in the selected reaction monitoring mode (SRM). The dwell time for all analytes was 5.0 ms. The other analyte-dependent parameters including the collision energy (CE), declustering potential (DP), entrance potential (EP) and collision cell exit potential (CXP) were optimized by infusing an aloin A and aloe emodin tuning solution using the built-in syringe pump at a flow rate of 10 μL min⁻¹ (Table 2).

Preparation of standard solutions

The primary stock solution of aloin A was prepared in 50% MeOH (v/v). Aloe emodin was prepared in DMSO. The concentrations were approximately 100 μg mL⁻¹ for both analytes. The stock solutions were stored at –18 °C. The secondary stock solutions at a concentration of 10 μg mL⁻¹ for both analytes were prepared in 50% MeOH from the primary stock solutions. The individual standard calibration solutions were prepared in 40% ACN/60% water (v/v) from the secondary stock solutions. The concentrations of the standard

solutions ranged from 5.0 to 1000 ng mL⁻¹ for aloin A and from 1.0 to 500 ng mL⁻¹ for aloe emodin.

Stability studies

Two kinds of stability studies for standards were performed. (1) Autosampler stability: a set of standards was placed in the UPLC autosampler tray (at 5 °C) and injected on days 1, 2, 3, 5 and 6. Calibration was performed for each day. (2) Short-term freezer stability: on day 1, one set of standards was prepared from the standard solutions and was consecutively injected three times. The standard solutions were stored in a freezer (–18 °C). On day 11 and day 33, the standard solutions were moved from the freezer. After equilibrating at room temperature, one set of standards was prepared from the standard solutions and was consecutively injected three times.

Sample preparation

For capsule and soft gel products, 12 capsules or 12 soft gels were opened and the contents were combined separately into 6 mL vials. The contents were mixed well to reduce variations due to sample inhomogeneity.⁴⁵ One of the products, labeled as tabs by the manufacturer, consisted of thin sheets of approximately 2 cm × 3 cm. Twelve of the tabs were cut into about 5 mm × 5 mm pieces for analysis.

For capsule, soft gel and tab products, a 10–20 mg test portion was weighed in a 50 mL centrifuge tube and 10.0 mL of extraction solution (40 : 60 ACN : H₂O) was added to the tube. The tube was vortexed for one minute. After sonicating for ten minutes, the tube was vortexed for another minute to rinse the wall followed by centrifugation for ten minutes at 5000 revolutions per minute (equivalent to 4500g) and filtration of the supernatant through a 0.2 μm PTFE filter (Pall Life Sciences, Port Washington, NY) into an autosampler vial for LC/MS-MS injection.

A 2.0 mL test portion of liquid samples was prepared in 8.0 mL of extraction solution (40 : 60 ACN : H₂O) contained in a 50 mL centrifuge tube. The remainder of the extraction procedure was the same as that for solid samples.

Since *Aloe vera* containing products varied in form and content of analytes, each product was first screened to determine whether matrix effects (signal suppression or enhancement) existed or whether the sample needed to be diluted to fit into the calibration range. If a sample did show serious matrix effects, the sample was further diluted as needed using the extraction solution (40 : 60 ACN : H₂O). The apparent recovery was used to evaluate the matrix effect and decide which samples needed to be diluted. If the recovery was lower than 60% or greater than 130%,

Table 2 Chemical formulae, molecular weights and SRM acquisition parameters^a

Analytes	Chemical formula	Average mass	Monoisotopic mass	SRM transition A	SRM transition B	DP (V)	EP (V)	CE (V)	CXP (V)
Aloin A	C ₂₁ H ₂₂ O ₉	418.399	418.126	417.2 → 297.0	417.2 → 268.1	–210	–10	–35	–10
Aloe emodin	C ₁₅ H ₁₀ O ₅	270.241	270.053	269.0 → 240.1	269.0 → 223.1	–140	–10	–35	–10

^a Note: SRM transition A was used for quantitation and SRM transition B was used for confirmation. CE = collision energy; DP = declustering potential; EP = entrance potential; CXP = collision cell exit potential.

the sample matrix was considered to cause serious ion suppression or enhancement, respectively.

Extraction efficiency study

For the extraction efficiency study, one capsule product was consecutively extracted four times using the procedure for capsule products. The four individual extracts were injected into the LC-MS/MS to quantify the analytes.

Recovery study

For the recovery study, known amounts of analytes (1000 ng of each analyte) were added to the sample before extraction and the same procedure as above was used to extract the spiked sample. Since some samples contain analytes and some samples lack analytes, the recoveries for these two types of samples were calculated using different formulae. For samples lacking analytes, the recoveries were calculated by the following formula:

$$\text{Recovery (\%)} = \left[\frac{\text{detected amount}}{\text{spiked amount}} \right] \times 100$$

For samples containing the analytes, the recoveries were calculated as follows:

$$\text{Recovery (\%)} = \left[\frac{\text{total amount} - \text{amount in sample}}{\text{spiked amount}} \right] \times 100$$

where the total amount is the amount determined after spiking; amount in sample is the amount without spiking and spiked amount is the known spiked amount.

Wash step between sample injections to eliminate carryover

To prevent excessive column carryover, the UPLC-MS/MS system was washed with six injections of a blank solution (40 : 60 ACN : H₂O) using a one-minute program followed by one injection of the blank solution using a three-minute "equilibration" gradient elution program. A mixture of 10 : 90 mobile phase A : mobile phase B at a flow rate of 0.5 mL min⁻¹ was used for the one-minute program. The parameters for the 3-minute equilibration gradient elution program were as follows: 0–2.0 min from 90% B to 40% B, 2.0–3.0 min 40 B% at 0.50 mL min⁻¹.

Results and discussion

Initially, a UPLC-UV/PDA method using the gradient program we described for elution of aloin A and aloe emodin was attempted to detect the analytes. Different wavelengths including those used in the published HPLC-UV methods^{22–25} were tried on our UPLC-UV system. It was found that the UPLC-UV results (not shown) for some samples were much higher (>150.0%) than those obtained by the UPLC-MS/MS method and some interfering peaks could contribute to the analyte peaks of interest by UPLC-UV. The interferences could be avoided by carefully selecting different wavelengths for individual analytes in different samples. However, no single wavelength worked for aloin A and aloe emodin in all samples under these conditions.

Optimization of the mass spectrometer

ESI in the negative ionization mode resulted in the best sensitivity and the mass spectrometer was set in product-ion-scan mode by selecting the corresponding [M – H][–] ion as the precursor ion for each analyte. The most abundant product ion observed for aloin A was at *m/z* 297, which resulted from the loss of [C₄H₈O₄][–]. The most abundant product ion observed for aloe emodin was at *m/z* 240, which resulted from the loss of [CHO][–].³⁹ Finally, the specific transitions selected for each analyte were: *m/z* 417.2 → 297.0 for aloin A and *m/z* 269.0 → 240.1 for aloe emodin. The transitions of 417.2 → 268.1, corresponding to the loss of [C₅H₉O₅][–], and 269.0 → 223.1, corresponding to the loss of [CH₂O₂][–], were used as confirmation for aloin A and aloe emodin, respectively.

Optimization of LC conditions

Different C₁₈ columns were tested. It was found that the use of a 1.7 μm C₁₈ column resulted in a shorter run time and sharper peaks. Although both methanol and acetonitrile generated good separation for the analytes, MeOH was preferred for this study due to its lower cost. The mixture of methanol and water caused a much higher back pressure (about 11 000 psi at a flow rate of 0.5 mL min⁻¹) than did a mixture of acetonitrile and water. This back pressure was within the allowable pressure with our UPLC system since the maximum allowable pressure was 15 000 psi. Representative chromatograms of aloin A and aloe emodin are shown in Fig. 2.

Linearity

An eight-point calibration curve using the peak area was created for concentrations ranging from 5.0–1000 ng mL⁻¹ for aloin A with *r* ≥ 0.995. An eight-point calibration curve using the peak area was generated for concentrations ranging from 1–500 ng mL⁻¹ for aloe emodin with *r* ≥ 0.995. To obtain the best accuracy, a weighting factor, 1/*x*², was applied for calibration, where *x* represents the concentration of analytes.

Intraday validation, accuracy and precision

The intraday validation data are summarized in Tables 3 and 4, respectively. For the six consecutive injections, both aloin A and

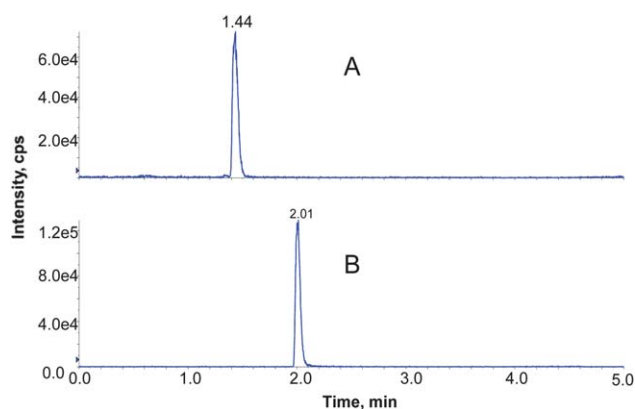


Fig. 2 Representative chromatograms of aloin A and aloe emodin standards from LC-MS/MS analysis. (A) aloin A and (B) aloe emodin.

Table 3 Intraday validation of standard calibration curves^a

Set	Aloin A			Aloe emodin		
	Slope	Inter.	<i>r</i>	Slope	Inter.	<i>r</i>
1	74 600	1720	0.9997	19 200	−508	0.9997
2	74 500	4130	0.9996	19 200	2440	0.9997
3	74 600	3960	0.9992	19 300	1570	0.9997
4	75 900	2530	0.9995	19 800	2570	0.9997
5	77 000	−70	0.9996	20 000	2610	0.9996
6	76 600	2470	0.9977	20 300	3690	0.9997
Mean	75 533	2457	0.9992	19 633	2062	0.9997
RSD%	1.48	63.03	0.08	2.38	69.26	0.00

^a Note: Inter. = intercept.

aloe emodin show high precision with an RSD < 3% for both slope and correlation coefficient (Table 3). The accuracies for the lower limit of quantitation (LLOQ), medium concentration and upper limit of quantitation (ULOQ) are listed in Table 4. The accuracies ranged from 93.0% to 104.0% for aloin A and 94.7% to 103.0% for aloe emodin. The precision for both analytes was less than 2.0%.

Stability studies

Two kinds of stability results are listed in Tables 5 and 6, respectively. The standards were stable in the autosampler (5 °C) for at least six days and they were stable in a freezer (−18 °C) for at least 33 days.

Carryover

A serious carryover for aloe emodin was observed after one injection of a high concentration (500 ng mL^{−1}). The carryover was substantially eliminated by using the wash steps described in the experimental section.

Extraction efficiency

The extraction efficiency results using spiked and un-spiked sample 8 are summarized in Table 7. A single extraction is sufficient to recover at least 98.0% of both analytes.

Table 4 Intraday accuracy and precision of individual standards in the calibration curve^a

Set	Aloin A						Aloe emodin					
	5.00 ng mL ^{−1}		50.00 ng mL ^{−1}		1000.00 ng mL ^{−1}		1.00 ng mL ^{−1}		20.00 ng mL ^{−1}		500.00 ng mL ^{−1}	
	Det. con., ng mL ^{−1}	Accu.%	Det. con., ng mL ^{−1}	Accu.%	Det. con., ng mL ^{−1}	Accu.%	Det. con., ng mL ^{−1}	Accu.%	Det. con., ng mL ^{−1}	Accu.%	Det. con., ng mL ^{−1}	Accu.%
1	4.96	99.20	49.80	99.70	952.00	95.20	1.01	101.00	20.60	103.00	474.00	94.80
2	4.91	98.20	51.10	102.00	938.00	93.80	1.00	99.90	20.30	102.00	476.00	95.30
3	4.98	99.60	51.00	102.00	957.00	95.70	1.00	99.70	20.10	101.00	474.00	94.70
4	4.94	98.70	52.00	104.00	945.00	94.50	0.99	98.90	20.40	102.00	479.00	95.80
5	4.95	99.00	50.40	101.00	942.00	94.20	1.02	102.00	20.50	103.00	479.00	95.90
6	4.99	99.70	51.50	103.00	930.00	93.00	1.00	99.70	20.40	102.00	477.00	95.40
Mean	4.96	99.07	50.97	101.95	944.00	94.40	1.00	100.20	20.38	102.17	476.50	95.32
RSD%	0.6	0.6	1.5	1.5	1.0	1.0	1.1	1.1	0.8	0.7	0.5	0.5

^a Note: concentrations in bold are theoretical concentrations. Det. Con. = determined concentration; Accu. = accuracy.**Table 5** Interday validation of the standard calibration curve^a

Day	Aloin A			Aloe emodin		
	Slope	Inter.	<i>r</i>	Slope	Inter.	<i>r</i>
1	12 000	3550	0.9993	19 100	731	0.9997
2	11 800	1930	0.9995	17 300	1200	0.9901
3	10 800	3440	0.9994	22 700	3260	0.9995
5	11 600	7980	0.9990	21 500	4310	0.9990
6	13 400	553	0.9990	24 800	1620	0.9991
Mean	11 920	3491	0.9992	21 080	2224	0.9975
RSD%	7.92	80.04	0.02	14.00	67.69	0.41

^a Note: the same set of standards were kept in the autosampler at 5 °C.

Matrix effects

One limitation of HPLC-ESI-MS/MS is the susceptibility to matrix effects.⁴⁶ Matrix effects are the alteration of ionization efficiency of target analytes in the presence of co-eluting compounds in the same matrix which can suppress or enhance the ESI signal of the target analyte when using external calibration (standard in solvent).^{46–51} Evaluating and minimizing ion suppression and enhancement are important considerations during method development. In this study, known amounts of native analytes were spiked into the samples and the recoveries were calculated to evaluate the matrix effects.

Several techniques to minimize matrix effects include: (1) use of stable isotopically labeled internal standards; (2) varying sample preparations, such as diluting the sample solution; (3) reducing the injection volume; (4) use of the method of standard additions; (5) use of matrix-matched standards; (6) use of structurally similar unlabelled compounds that elute close to the

Table 6 Storage stability of calibration standards (*n* = 3; −18 °C)

Day	Aloin A			Aloe emodin		
	Slope	Inter.	<i>r</i>	Slope	Inter.	<i>r</i>
1	12 667	1854	0.9993	20 033	2483	0.9996
10	12 150	2997	0.9993	20 400	2143	0.9994
33	12 300	11 507	0.9993	19 950	4350	0.9992
Mean	12 372	5453	0.9993	20 128	2992	0.9994
RSD%	2.15	96.73	0.00	1.19	39.71	0.02

Table 7 Extraction efficiency for un-spiked and spiked capsule sample 8^a

Samples	Ext.	Aloin A			Aloe emodin		
		Det. amt, ng	RSD	Ext. effi.	Det. amt, ng	RSD	Ext. effi.
Unspk. sam.	1 st ext.	1430.0	0.7%	99.1%	81.4	0.9%	100.0%
	2 nd ext.	13.1	9.5%	0.9%	ND	NA	NA
	3 rd ext.	ND	NA	NA	ND	NA	NA
	4 th ext.	ND	NA	NA	ND	NA	NA
Spk. sam.	1 st ext.	2167.0	0.7%	99.1%	838.3	1.1%	98.7%
	2 nd ext.	19.2	4.3%	0.9%	11.4	6.1%	1.3%
	3 rd ext.	ND	NA	NA	ND	NA	NA
	4 th ext.	ND	NA	NA	ND	NA	NA

^a Note: ND = none detected; NA = not applicable; Unspk. sam. = un-spiked sample; Spk. sam. = spiked sample; Ext. = extraction; Det. amt. = determined amount; Eff. = efficiency.

compounds of interest as internal standards; and (7) varying chromatographic conditions.^{46–51}

Although the use of stable isotopically labeled standards is the preferred technique to compensate for matrix effects, appropriate compounds for this study were not commercially available. The method of standard additions was not practical for this study because some of the samples contained high concentration of the analytes. The structurally similar analogue, emodin,⁴⁰ was tested as an internal standard at the beginning of this study. Our results showed that the emodin standard contains quantifiable amounts of aloe emodin. Emodin was also reported in natural products.⁵² Therefore, emodin was not a suitable internal standard for this study. In addition, matrix blanks to prepare matrix-matched standards were not available because the products varied in form and content of analytes.

In this study, matrix effects were minimized by simple dilution. Aloin A and aloe emodin have similar structures, but each sample matrix had a different matrix effect on the respective analytes. For instance, some samples showed little matrix effect on the analytes. Others showed serious ion suppression with an apparent recovery <40% and one showed a significant signal enhancement with an apparent recovery >170% only for aloin A. A similar phenomenon has also been observed.⁵³ By first evaluating the matrix effects, followed by diluting the test

portion of the solution, we found that the results were acceptable for both analytes with recoveries ranging from 74.2% to 117.5%. Recovery for most samples ranged from 85.0% to 110.0% (Table 8).

Also shown in Table 8 are the dilution factors used to minimize the matrix effects. For some samples, no dilution was needed; six samples needed 5-fold dilution. It should be pointed out that samples 10 and 11 contained very high concentrations of the analytes, so 100-fold and 2500-fold dilutions were needed in order to fit into the calibration range.

Optimization of sample preparation

Initially, methanol–water was used to extract the samples and 0.2 µm nylon filters were used to filter the extracts. It was found that the use of the nylon filter resulted in a lower recovery (about 50%) for aloe emodin. To eliminate this loss for aloe emodin, a PTFE filter was used. However, the use of a PTFE filter made the filtration more difficult due to back pressure. To improve the filtration efficiency, different ratios of ACN and water were tested. The use of 40 : 60 ACN : H₂O yielded a higher recovery and better peak shapes for the analytes with a lower filtration back pressure.

Table 8 Results for products labeled to contain *Aloe vera*^a

Sample # (n = 3)	Form	Aloin A				Aloe emodin				Dilution factor
		Det. µg g ⁻¹	Det. RSD%	Rec.%	Rec. RSD%	Det. µg g ⁻¹	Det. RSD%	Rec.%	Rec. RSD%	
1	Liq.	ND	NA	97.1	3.7	ND	NA	108.3	0.7	1
2	Liq.	ND	NA	100.3	1.5	ND	NA	95.1	3.2	1
3	Liq.	ND	NA	97.6	0.7	ND	NA	89.3	3.9	5
4	Liq.	ND	NA	101.4	4.1	0.2	1.5	90.4	1.0	5
5	Liq.	ND	NA	97.9	1.0	ND	NA	96.9	5.0	5
6	Liq.	ND	NA	96.8	2.5	0.1	3.0	96.4	2.9	5
7	Liq.	ND	NA	100.5	2.9	8.7	2.1	87.8	8.1	5
8	Cap.	89.2	3.2	94.6	2.7	5.2	6.3	78.1	3.4	1
9	Cap.	21.2	2.3	92.8	3.2	24.2	5.5	74.2	1.8	1
10	Cap.	35100	3.9	NS	NA	453.0	6.2	100.6	21.5	100
11	Cap.	160000	5.6	NS	NA	3810	9.6	NS	NA	2500
12	S. gel	ND	NA	89.3	3.4	ND	NA	84.2	7.8	1
13	S. gel	ND	NA	94.4	5.3	ND	NA	75.7	4.2	1
14	Tab	ND	NA	117.5	2.8	ND	NA	106.4	2.0	5

^a Note: Det. = determined; Rec. = recovery; ND = none detected; NA = not applicable; NS = not suitable due to high concentration; Liq. = liquid; Cap. = capsule; S. gel = soft gel.

Aloe product survey with recovery

Aloe vera products are available in various formulations including powder, liquids, tabs and capsules (Table 1). The claims made on the product labels include improvement in immune, digestive and gastrointestinal function and response.

In this study, fourteen aloe products (7 liquid samples, 4 capsule samples, 2 soft gel samples and 1 tab sample) were analyzed using the UPLC-MS/MS method. The results are summarized in Table 8. A representative chromatogram of aloe product #9 is shown in Fig. 3. As can be seen in Fig. 3A and B, there were no serious interference peaks for aloe emodin. A large peak appeared in front of aloin A. This peak is most likely aloin B because both peaks have the same peak height ratios of the quantitation transition A (417.2 $\rightarrow$ 296.7) to that of the confirmation transition 417.2 $\rightarrow$ 268.1. The two diastereoisomers, aloin A and aloin B, were also reported by other studies.^{12,22,25,40,44} Since the peaks for aloin A and aloin B could not be baseline-separated, the peak for aloin A was manually integrated for all samples. Pure standard aloin B was not available. Based on RSD values (all less than 9.6%) and recoveries (89.3%–117.5%) for aloin A, manually integrated results were acceptable.

The amount of aloin A and aloe emodin in the commercial products ranged from none detected to 160.0 mg g⁻¹, and from none detected to 3.8 mg g⁻¹, respectively. Other studies showed that the amount of aloin A in commercial aloe-based products ranged from none detected to 10.2 mg g⁻¹⁴⁴ and from none detected to 0.305 mg g⁻¹.⁴⁰ The amount of aloe emodin in aloe-based products ranged from none detected to 0.114 mg g⁻¹.⁴⁰ The recoveries for aloin A and aloe emodin ranged from 89.3% to 117.5%, and 74.2% to 108.3%, respectively, in the matrices tested.

Table 8 presents the results for the analysis of 14 commercial products labeled to contain *Aloe vera*. The levels of aloin A and aloe emodin in these products varied widely. In general, the labeling of the products provided limited information on the source (e.g., whole leaf, inner leaf, latex) or the processing method (e.g., unfiltered, charcoal filtered) of the *Aloe vera* used. Therefore, it is difficult to attribute the levels of aloin A and aloe emodin to specific sourcing and processing methods for many of the products. An association was noted between the

physical form of the products and the amount of aloin A and aloe emodin found. No aloin A was found in any of the liquid products (products 1–7). The labels on products 2, 3 and 6 stated that the *Aloe vera* used had been processed to remove aloe emodin and aloin. Consistent with their labeling, products 2 and 3 were found to contain no detectable aloin A or aloe emodin. However, aloe emodin was found in product 6. Product 7 contained the highest level of aloe emodin observed for the analyzed liquid products. The relatively high level of aloe emodin in product 7 may reflect both inadequate processing to remove anthraquinones and the concentrated form of *Aloe vera* present in this product. Both aloin A and aloe emodin were found in all of the capsules analyzed (products 8–11). Products 8 and 9 contained the lowest levels of aloin A and aloe emodin found in capsules. Product 8 was labeled to contain the inner leaf gel of *Aloe barbadensis* obtained by hand filleting the leaf. Product 9 was labeled to contain the gel, skin and sap of aloe which had been treated to remove aloin. Higher levels of aloin A and aloe emodin were found in products 10 and 11. Product 10 was labeled to contain both the latex of *Aloe ferox* and the leaf of *Aloe barbadensis*. As previously noted, anthraquinones are found in the latex of aloe. Product 11 contained the highest level of aloin A and aloe emodin. The label of product 11 states that this product contains the dried resin of *Aloe ferox* and is used as a laxative. This intended use is consistent with the levels of anthraquinones found in product 11. None of the products in soft gel (products 12 and 13) or tab (product 14) form were found to contain either aloin A or aloe emodin. These products were labeled to contain *Aloe vera* extracts. The absence of aloin A and aloe emodin suggests that these extracts had been processed to remove these anthraquinones. The results of this product analysis illustrate a variety of sources and processing methods used for *Aloe vera* and the effects on levels of anthraquinones in the final product.

The amount of aloin A in all capsule products (8 through 11) exceed the 10 $\mu\text{g g}^{-1}$ (ppm) limit which was established for aloin in products for oral consumption by the International Aloe Science Council.⁹

Based on the data in Table 8 and serving size information, sample 11 which contains the highest amount of aloin A was used to calculate the amount of aloin A and aloe emodin that could be consumed per day as follows:

Serving size/suggested use: 1 capsule (775 mg = 0.775 g).

For aloin A: $0.775 \text{ g} \times 160\,000 \mu\text{g g}^{-1} = 124\,000 \mu\text{g} = 124.0 \text{ mg}$.

For aloe emodin: $0.775 \text{ g} \times 3810 \mu\text{g g}^{-1} = 2952 \mu\text{g} = 3.0 \text{ mg}$.

i.e., 124.0 mg aloin A and 3.0 mg aloe emodin would be consumed per day.

Conclusion

Aloe has been widely used for manufacturing food products, beverages, cosmetics and pharmaceuticals due to its aromatic properties, bitter taste, the cathartic activity of anthraquinones and other pharmacological activities. A rapid and sensitive analytical method for quantifying aloin A and aloe emodin using UPLC-ESI-MS/MS has been developed, evaluated and successfully applied to consumer products labeled to contain *Aloe vera*. To our knowledge, this is the first study of the use of a UPLC system with a 1.7 μm C₁₈ column for separation, and tandem

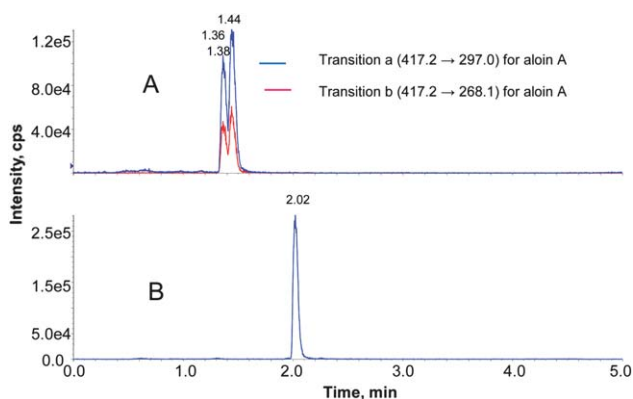


Fig. 3 Representative chromatograms of aloin A and aloe emodin for capsule sample 9 from LC-MS/MS analysis. (A) aloin A and (B) aloe emodin.

mass spectrometry (MS/MS) for quantitation and confirmation of aloin A and aloe emodin in different aloe products. This method should be a valuable tool for determining aloin A and aloe emodin in consumer products.

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